

A FORWARD MODELING PROMPTS

Prompts for forward modeling zero-shot are in Table 5, and prompts for forward CoT are in Table 6.

Table 5: Zero-shot prompting for forward modeling paradigms

LLMs predict individual human participant’s choice:

A person is presented with two gambling machines, and makes a choice between the machines with the goal of maximizing the amount of dollars received.

The person will get one reward from the machine they choose. A fixed proportion of 10% of this value will be paid to the participant as a performance bonus.

If the reward is negative, their bonus is set to \$0.

Machine A: {}

Machine B: {}

Which machine does the person choose?

Do not provide any explanation, only answer with A or B:

LLMs predict the human choice distribution:

{ } people are presented with two gambling machines, and each person makes a choice between the machines with the goal of maximizing the amount of dollars received.

Each person will get one reward from the machine they choose. A fixed proportion of 10% of this value will be paid to the participant as a performance bonus.

If the reward is negative, their bonus is set to \$0.

Machine A: {}

Machine B: {}

How many people choose Machine A?

How many people choose Machine B?

Please only provide the percentage of people who choose Machine A and Machine B in the json format.

LLMs act as human participant:

There are two gambling machines, A and B. You need to make a choice between the machines with the goal of maximizing the amount of dollars received.

You will get one reward from the machine that you choose. A fixed proportion of 10% of this value will be paid to you as a performance bonus.

If the reward is negative, your bonus is set to \$0.

Machine A: {}

Machine B: {}

Which machine do you choose?

Do not provide any explanation, only answer with A or B:

B FORWARD MODELING RESULTS WITH DIFFERENT PROMPTS

To investigate how LLMs estimate human behavior for the overall sample sizes, we asked LLMs to predict the probability distribution of human choices between gamble machine A and gamble machine B. We observed that estimate the probability of the overall decisions mitigates the strong correlation with the maximum expected value of each machine while not improving the correlation with actual human behaviors. Particularly for zero-shot prompting, both closed-source and open-source models in Table 7 show a drop in correlation values between 14% and 25% compared to the zero-shot prompting results in Table 1.

We also observed that even when LLMs are asked to act as human participants in making decisions, their outcomes remain consistently more rational than actual human behavior under CoT prompting. Table 10 shows the results of LLMs’ decisions as individual human participants compared to the maximum expected value. Both GPT-4-Turbo and GPT-4o exhibit a high correlation with the maximum expected value, each with a correlation coefficient of 0.91 and 0.92 and the MSE of 0.045 and 0.037, while still maintaining a moderate correlation with actual human behavior, as shown in Table 9.

Table 6: CoT prompting for forward modeling paradigms

LLMs predict individual human participant’s choice:

A person is presented with two gambling machines, and makes a choice between the machines with the goal of maximizing the amount of dollars received.

The person will get one reward from the machine they choose. A fixed proportion of 10% of this value will be paid to the participant as a performance bonus.
If the reward is negative, their bonus is set to \$0.

Machine A: {}
Machine B: {}

Which machine does the person choose?
Let’s think step by step before answering with A or B:

LLMs predict the human choice distribution:

{ } people are presented with two gambling machines, and each person makes a choice between the machines with the goal of maximizing the amount of dollars received.
Each person will get one reward from the machine they choose. A fixed proportion of 10% of this value will be paid to the participant as a performance bonus.
If the reward is negative, their bonus is set to \$0.

Machine A: {}
Machine B: {}

How many people choose Machine A?
How many people choose Machine B?

Let’s think step by step before providing the final output.
Please provide the percentage of people who choose Machine A and Machine B in the json format.

LLMs act as human participant:

There are two gambling machines, A and B. You need to make a choice between the machines with the goal of maximizing the amount of dollars received.
You will get one reward from the machine that you choose. A fixed proportion of 10% of this value will be paid to you as a performance bonus.
If the reward is negative, your bonus is set to \$0.

Machine A: {}
Machine B: {}

Which machine do you choose?
Let’s think step by step before answering with A or B:

Table 7: The correlation between LLMs predicting the human choice distribution based on the aggregate sample size of participants and the actual human choice.

		Llama3-8B	Llama3-70B	GPT-4-Turbo	GPT-4o	Humans
Zero-shot	Spearman correlation	0.1045	0.2827	0.4812	0.6156	/
	Pearson correlation	0.1032	0.2904	0.4830	0.6112	/
	MSE	0.2811	0.2668	0.1951	0.1633	/
CoT	Spearman correlation	0.1799	0.1046	0.6208	0.5825	/
	Pearson correlation	0.1783	0.0992	0.6308	0.6012	/
	MSE	0.2615	0.2954	0.1202	0.1282	/

C MODEL CORRELATIONS FOR FORWARD MODELING

We provide the full correlation results for the three forward modeling paradigms for Llama3-8B, Llama3-70B, GPT-4-Turbo (0125-preview), and GPT-4o in Figure 2. Compared to the correlations in zero-shot prompting, CoT prompting shows a higher degree of correlation across all four LLMs.

D COMPARING LLMs TO A WIDER RANGE OF BEHAVIORAL MODELS

Our results clearly show that chain-of-thought results in LLM responses that align closely with expected value. To try to understand whether there was a systematic pattern in the responses of the

Table 8: The correlation between LLMs predicting the human choice distribution based on the aggregate sample size of participants and the maximum expected value.

		Llama3-8B	Llama3-70B	GPT-4-Turbo	GPT-4o	Humans
Zero-shot	Spearman correlation	0.0426	0.1688	0.3380	0.1741	0.4835
	Pearson correlation	0.0426	0.1688	0.3380	0.1741	0.4835
	MSE	0.4752	0.4361	0.3465	0.4527	0.2580
CoT	Spearman correlation	0.2025	0.8406	0.8458	0.8518	0.4835
	Pearson correlation	0.2025	0.8406	0.8458	0.8518	0.4835
	MSE	0.3978	0.0807	0.0726	0.0702	0.2580

Table 9: The correlation between LLMs acting as a human participant to make choice and the actual human choice.

		Llama3-8B	Llama3-70B	GPT-4-Turbo	GPT-4o	Humans
Zero-shot	Spearman correlation	0.4047	0.4528	0.5841	0.4617	/
	Pearson correlation	0.4068	0.4559	0.5667	0.4565	/
	MSE	0.0920	0.1372	0.1414	0.2031	/
CoT	Spearman correlation	0.4597	0.6223	0.6153	0.6074	/
	Pearson correlation	0.4600	0.6165	0.6115	0.6030	/
	MSE	0.1972	0.1621	0.1640	0.1659	/

Table 10: The correlation between LLMs acting as a human participant to make choice and the maximum expected value.

		Llama3-8B	Llama3-70B	GPT-4-Turbo	GPT-4o	Humans
Zero-shot	Spearman correlation	0.1774	0.3130	0.4190	0.2944	0.4835
	Pearson correlation	0.1730	0.3069	0.4053	0.2944	0.4835
	MSE	0.2888	0.2980	0.2729	0.3536	0.2580
CoT	Spearman correlation	0.5155	0.8353	0.9100	0.9255	0.4835
	Pearson correlation	0.5155	0.8353	0.9100	0.9255	0.4835
	MSE	0.2492	0.0836	0.0450	0.0372	0.2580

zero-shot models, we fit 18 choice models from the behavioral sciences to the output of GPT-4 using the zero-shot individual choice prompt. The set of models was based on those used in Peterson et al. (2021), where they are described in more detail together with references to the original paper.

In this context, each model represents a hypothesis about the LLM’s beliefs about human behavior. These included Heuristic models (He et al., 2022), wherein people are thought to employ mental shortcuts to make decisions, Counterfactual models (He et al., 2022), which appeal to constructs like regret and disappointment, and Subjective Expected Utility models (He et al., 2022), which assume that quantities involved (money and probability) are perceived or otherwise treated subjectively. This latter, third category contains many of the most influential models—Expected Utility Theory (EU) and Prospect Theory (PT)—as well as the model proposed by Peterson et al. (2021) which is called Mixture of Theories (MOT).

Table 11 shows the results. Models in the top two sections of the table (Heuristic and Counterfactual) provided strictly inferior fits compared to Subjective Expected Utility models in the third section. Among those in the third section, Expected Value provided the worst fit. Expected Utility was notably better, suggesting that GPT-4 correctly assumes that people do not treat the value of money objectively / linearly. Prospect Theory improved this score slightly through the incorporation of a subjective probability weighting function, but that fitted function was largely linear, suggesting that GPT-4 incorrectly assumes that people do not treat probabilities subjectively. Lastly, MOT provided the best fit to the inferences of GPT-4. In previous work, MOT also provides the best fit to human data, but the fitted parameters are different (Peterson et al., 2021). When fitted directly to `choices13k`, MOT learns a mixture of two utility functions (e.g., like the one in Expected Utility) and two probability